



HEALTHCONNECT CLINIC — WEEK 5 ANALYTICS REPORT

AnalystLab Africa Experience Lab

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Week 4 → Week 5 Transition

In Week 4, I reviewed the HealthConnect Appointment Dataset (5,000 records, 18 variables) and conducted an initial data quality assessment, finding the dataset largely clean with only minor, logically explainable missing values in `distance_to_clinic_km` (1.8%), `waiting_time_minutes` (1.2%), and `reminder_channel` (fully explained by unsent reminders). I identified 7 business questions and proposed 5 KPIs — Overall No-Show Rate, No-Show Rate by Reminder Status/Channel, No-Show Rate by Previous No-Show History, Average Distance to Clinic, and No-Show Rate by Booking Lead Time — each tied to a specific question about what drives patient no-shows. The headline finding was a striking 48.46% no-show rate across the dataset. My proposed focus for Week 5 was to prepare the data, calculate these KPIs for real, and begin exploratory analysis starting with reminder effectiveness and previous no-show history.

Data Preparation Summary

Building on the data quality assessment conducted in Week 4, the following cleaning and preparation steps were applied to the HealthConnect Appointment Dataset in Power BI this week. The original dataset file was not modified — all preparation was performed within Power Query as part of the Power BI project.

1. Data Type Verification

All columns were reviewed to confirm correct data types on import. `booking_date` and `appointment_date` were already correctly recognized as Date type. Numeric fields (`age`, `booking_lead_days`, `previous_appointments`, `previous_no_shows`, `distance_to_clinic_km`, `waiting_time_minutes`) and categorical fields (`gender`, `age_group`, `appointment_type`, `appointment_day`, `appointment_time`, `reminder_sent`, `reminder_channel`, `appointment_outcome`) were also confirmed to carry appropriate types with no corrections needed.

2. Missing Value Treatment

- **`distance_to_clinic_km`** (90 missing values, 1.8%) and **`waiting_time_minutes`** (60 missing values, 1.2%) were imputed using the **median** value of each column — 8.7 km and 24 minutes respectively. The median was chosen over the mean because it is more resistant to distortion by outliers, making it a more representative estimate for the typical patient. This preserved all 5,000 records for analysis rather than excluding rows with incomplete data.
- **`reminder_channel`** (1,366 missing values, 27.3%) was deliberately left unmodified. Investigation confirmed every missing value in this column corresponds exactly to `reminder_sent = "No"` — meaning there is no reminder channel to record for these appointments. Filling these values would misrepresent the data rather than correct an error.

3. Verification

- Following all preparation steps, the dataset was confirmed to retain its full **5,000 records**, with no duplicate rows or duplicate appointment IDs, consistent with the findings from the Week 4 assessment.

Exploratory Data Analysis & KPI Findings

Five KPIs were selected from the candidates proposed in Week 4, calculated using DAX measures in Power BI, and visualized to investigate the drivers of patient no-shows. The overall baseline No-Show Rate across all 5,000 appointments was confirmed at **48.46%**, consistent with the Week 4 preliminary finding.

KPI 1 — No-Show Rate by Reminder Status and Channel

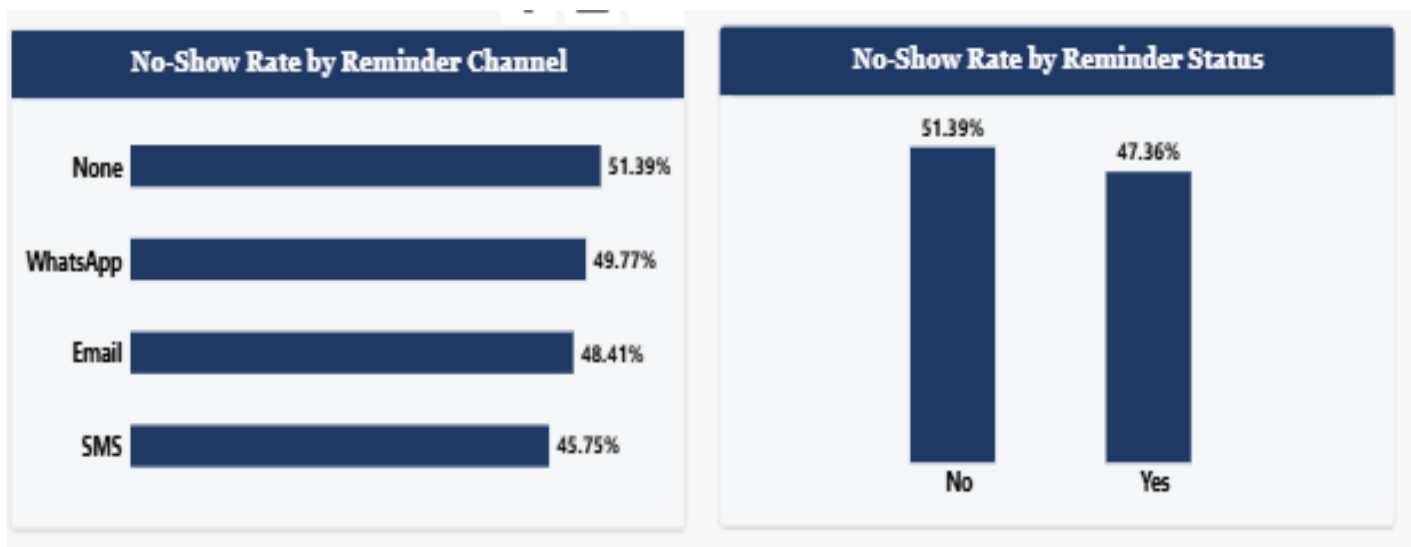
Definition: The percentage of appointments ending in a no-show, segmented by whether a reminder was sent and through which channel.

Why it matters: Tests whether the clinic's existing reminder system is effective, and identifies which channel performs best.

Business Question: Does sending a reminder — and which channel — reduce no-shows?

Result: Appointments with no reminder sent showed a 51.39% no-show rate, compared to 47.36% when a reminder was sent. By channel: SMS performed best (45.75%), followed by Email (48.41%) and WhatsApp (49.77%).

Interpretation: Reminders provide a modest but real improvement, and SMS is meaningfully more effective than WhatsApp or Email. This is a directly actionable finding — the clinic should prioritize SMS as its primary reminder channel.



KPI 2 — No-Show Rate by Previous No-Show History

Definition: The percentage of appointments ending in a no-show, segmented by the patient's count of prior no-shows.

Why it matters: Tests whether past behavior predicts future attendance, a common strong

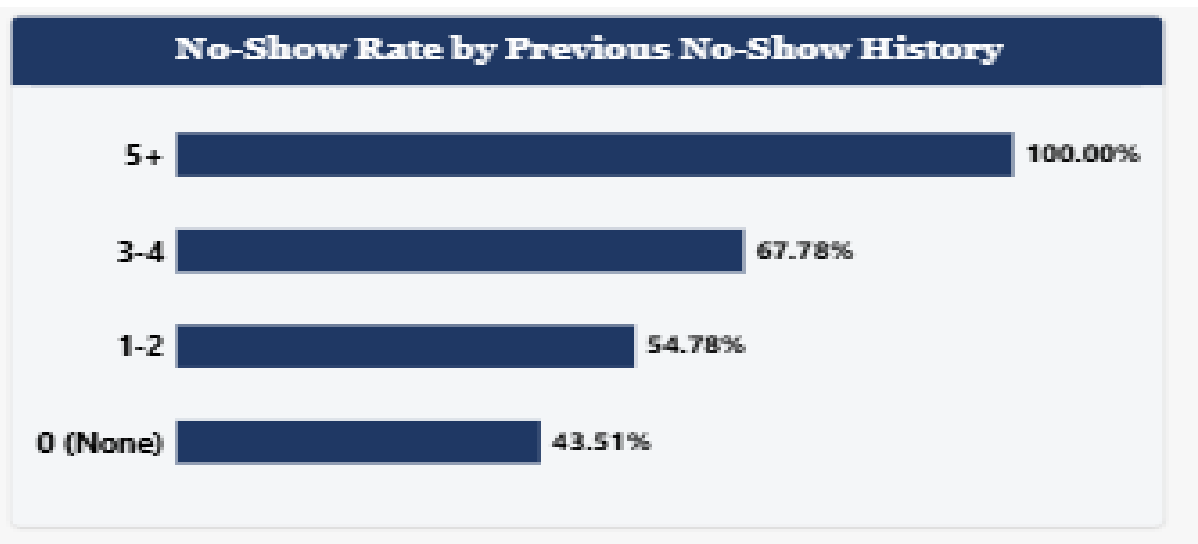
signal in appointment data.

Business Question: Does a patient's history of previous no-shows predict future no-shows?

Result: No-show rate rises steadily with prior history — 43.51% (no prior no-shows), 54.78% (1-2), 67.78% (3-4), and 100% (5+, though based on only 3 patients).

Grouping approach: Since previous_no_shows is a numeric count with a long tail (most patients have 0-2 prior no-shows, with very few having more), the values were grouped into 4 bands using a DAX calculated column: **0 (None)**, **1-2**, **3-4**, and **5+**. This grouping was chosen to keep each band meaningful in size while still isolating patients with escalating risk levels, rather than charting every individual count value (which would produce too many thin, hard-to-read bars).

Interpretation: Previous no-show history is a strong, consistent escalating risk signal (correlation 0.126), making it a valuable input for identifying high-risk patients proactively.



KPI 3 — Average Distance to Clinic (No-Show vs Attended)

Definition: The percentage of appointments ending in a no-show, segmented by distance band from the clinic.

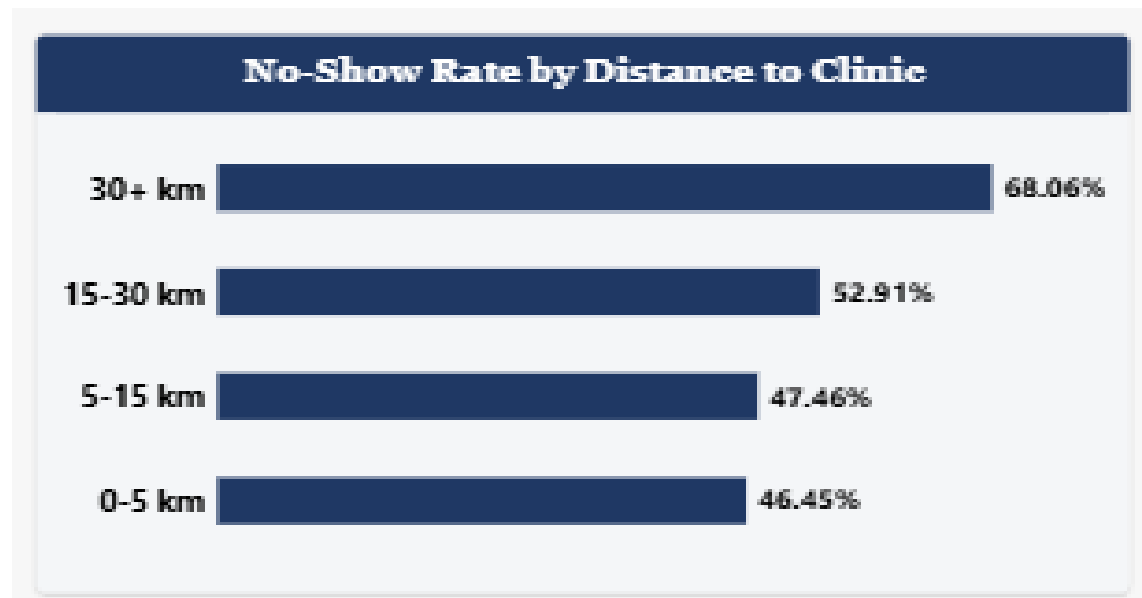
Why it matters: Tests whether travel burden is a practical barrier to attendance.

Business Question: Does distance to the clinic affect the likelihood of a no-show?

Grouping approach: distance_to_clinic_km was grouped into 4 bands — **0-5 km**, **5-15 km**, **15-30 km**, and **30+ km** — using a DAX calculated column. These bands were chosen to reflect realistic travel-difficulty tiers (short local trips, moderate commutes, longer journeys, and distant outliers), allowing the analysis to detect whether risk changes gradually or concentrates at an extreme.

Result: No-show rate is relatively flat for patients within 15km (46-47%), rising to 52.91% (15-30km) and sharply to 68.06% (30km+, though based on a small sample of 72 patients).

Interpretation: Distance is a weak overall predictor (correlation 0.061) but becomes meaningful at the extreme end, suggesting a targeted intervention (e.g., telehealth options) for distant patients rather than a universal distance-based policy.



KPI 4 — No-Show Rate by Booking Lead Time

Definition: The percentage of appointments ending in a no-show, segmented by the number of days between booking and the appointment date.

Why it matters: Tests whether longer gaps between booking and attending reduce commitment or increase forgetfulness.

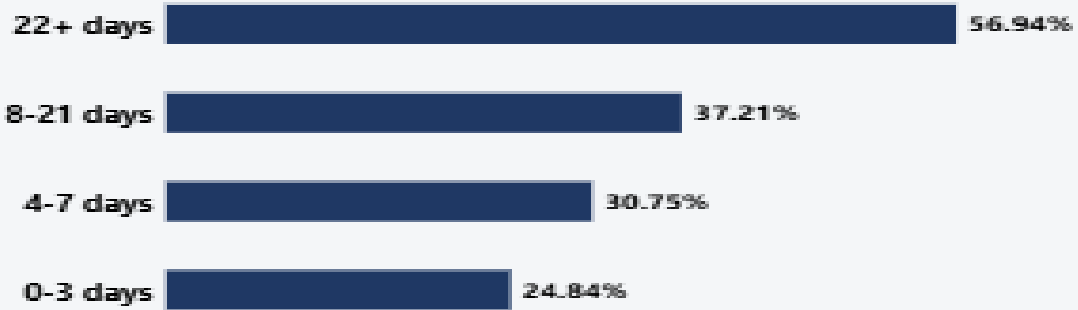
Business Question: Does the time between booking and the appointment affect attendance?

Grouping approach: booking_lead_days (ranging from 0 to 60 days) was grouped into 4 bands using a DAX calculated column — **0-3 days**, **4-7 days**, **8-21 days**, and **22+ days**. These bands were chosen to distinguish short-notice bookings from those made weeks in advance, capturing the point at which longer lead times begin to noticeably affect attendance rather than splitting the range arbitrarily.

Result: No-show rate climbs sharply with lead time — 24.84% (0-3 days), 30.75% (4-7 days), 37.21% (8-21 days), and 56.94% (22+ days).

Interpretation: This is the strongest individual predictor identified (correlation 0.272), suggesting the clinic should consider strategies for appointments booked far in advance, such as additional reminder touchpoints or confirmation calls closer to the date.

No-Show Rate by Booking Lead Time



Business Insights

1. **Booking lead time is the strongest single predictor of no-shows.** Appointments booked 22+ days in advance have a 56.94% no-show rate — more than double the 24.84% rate for appointments booked 0-3 days ahead. This suggests the gap between booking and attending is itself a risk factor, independent of who the patient is.
2. **Prior no-show history strongly predicts future no-shows.** Patients with no prior history no-show at 43.51%, rising steadily to 54.78% (1-2 prior no-shows), 67.78% (3-4), and 100% among the small group with 5+ prior no-shows — confirming past behavior is a meaningful, escalating risk signal.
3. **SMS reminders outperform other channels.** Patients reminded via SMS show a 45.75% no-show rate, compared to 48.41% (Email) and 49.77% (WhatsApp) — and all reminder types outperform sending no reminder at all (51.39%). This is a directly actionable finding: SMS should be prioritized as the primary reminder channel.
4. **Distance to the clinic matters mainly at the extreme end.** No-show rate stays relatively flat for patients within 15km (46-47%), but rises sharply for those living 15-30km away (52.91%) and especially 30km+ (68.06%) — though this furthest group is small (72 patients), suggesting a targeted rather than universal intervention is appropriate.
5. **The overall no-show rate (48.46%) is driven by a combination of controllable factors, not one dominant cause.** No single KPI alone fully explains the pattern (even the strongest, booking lead time, has a modest 0.272 correlation) — meaning an effective intervention strategy will likely need to combine several levers (shorter lead times where possible, SMS reminders, targeted outreach for high-risk-history patients) rather than relying on a single fix.

Cross-Track Collaboration

Track interacted with: Data Science

Information/output exchanged: The KPI findings from this week's analysis — particularly the relative strength of each factor (Booking Lead Time: correlation 0.272; Previous No-Show History: 0.126; Distance to Clinic: 0.061) — represent candidate features and their relative predictive value for the Data Science track's no-show prediction model, which was proposed in their Week 4 problem definition.

Why the interaction was relevant: The Data Science track needs to identify which input features are worth including in their baseline model. Since correlation strength was calculated as part of this week's Data Analytics work, sharing these findings directly informs their feature selection — rather than the Data Science track needing to independently re-derive which variables matter most.

What changed or improved as a result: This analysis suggests Booking Lead Time and Previous No-Show History should be prioritized as strong candidate features, while Distance to Clinic, though still relevant, may contribute less predictive value on its own and could be considered as a secondary or interaction feature rather than a primary driver. This gives the Data Science track an evidence-based starting point for feature selection rather than treating all proposed features as equally important.

Assumptions, Limitations, Risks and Dependencies

Assumptions

- **Still relevant:** The dataset remains assumed to be a representative, complete sample of appointment activity, and appointment_outcome is still assumed to be recorded accurately.
- **Still relevant:** The dataset's fictional/synthetic nature (noted in Week 4) continues to apply — findings remain illustrative rather than directly actionable for a real clinic.

Limitations

- **Resolved:** The missing values in distance_to_clinic_km (1.8%) and waiting_time_minutes (1.2%) identified in Week 4 have now been addressed through median imputation (8.7km and 24 minutes respectively), so this is no longer an open limitation for analysis using those fields.
- **Still relevant:** The dataset still lacks a "reason for no-show" field, meaning root-cause analysis remains limited to correlational patterns (e.g., *that* distance or lead time relates to no-shows) rather than *why* individual patients specifically missed their appointments.
- **New this week:** The correlation values calculated this week are notably modest even for the strongest factor (Booking Lead Time: 0.272) — this reveals a limitation not fully anticipated in Week 4: no single available variable strongly predicts no-shows in isolation, meaning any future predictive model will likely need multiple combined features rather than relying on one dominant driver.
- **New this week:** The "5+ Previous No-Shows" and "30+ km Distance" segments, while showing the most striking rates (100% and 68.06% respectively), are based on very small sample sizes (3 and 72 patients) — findings from these specific segments should be treated as directional rather than statistically robust.

Risks

- **Still relevant:** The risk noted in Week 4 — that over-relying on previous no-show history could create a self-fulfilling, potentially punitive pattern toward flagged patients — remains relevant now that this week's analysis has confirmed history is indeed a meaningful predictor (43.51% → 100% across bands).
- **New this week:** Since SMS reminders show the strongest effect among channel types, there is a risk of over-indexing on SMS without considering patient channel preference

or accessibility (e.g., some patients may not have reliable SMS access) — this should be tested further before becoming a blanket policy recommendation.

Dependencies

- **Confirmed and active:** As documented in the Cross-Track Collaboration section, this week's findings (particularly the relative strength of Booking Lead Time, Previous No-Show History, and Distance to Clinic) now feed directly into the Data Science track's feature selection process — this dependency, anticipated in Week 4, is now an active exchange rather than a proposed one.

Week 5 Project Summary — Data Analytics Track

1. What I Planned to Accomplish

Building on Week 4's foundation, I planned to prepare the dataset for real analysis, calculate the 5 proposed KPIs with actual numbers, build supporting visualizations, and produce evidence-based business insights — moving from planning into practical analysis.

2. What I Actually Completed

I cleaned the dataset in Power BI (correcting date field types, imputing missing `distance_to_clinic_km` and `waiting_time_minutes` values using the median), built 8 DAX measures including No-Show Rate, and calculated and visualized all 5 proposed KPIs: No-Show Rate by Reminder Status/Channel, by Previous No-Show History, by Distance to Clinic, and by Booking Lead Time. I produced 5 evidence-based business insights and completed one documented cross-track collaboration exchange with the Data Science track.

3. Key Findings

Booking Lead Time emerged as the strongest single predictor (correlation 0.272), with no-show rates rising from 24.84% (0-3 days) to 56.94% (22+ days). Previous No-Show History showed a similarly strong escalating pattern (43.51% → 100%). SMS reminders proved the most effective channel (45.75% no-show rate). Distance to Clinic showed a weaker but still meaningful effect concentrated among patients living 30km+ away (68.06%).

4. Major Challenges Encountered

The main challenge was deciding how to handle the two numeric fields with missing values — I chose median imputation over dropping rows or leaving blanks, since it preserved the full dataset while using a defensible, outlier-resistant estimate. I also had to think carefully about how to group continuous variables (distance, lead time, previous no-shows) into meaningful bands for clear visualization, rather than charting every individual value.

5. Important Decisions Made and Why

I chose to bin continuous numeric variables into categorical bands (e.g., "0-3 days," "4-7 days") rather than using scatter plots, since this made patterns immediately visible to a non-technical audience while still preserving the underlying trend. I also chose to

include correlation coefficients alongside percentage breakdowns, since percentages alone can visually suggest a stronger relationship than the data statistically supports (as seen with Distance to Clinic).

6. Changes to My Week 4 Approach

No significant changes — the Week 4 proposed approach (data prep, then KPI calculation by dimension, then synthesis) was followed as planned. One refinement: correlation strength was added as a supporting metric alongside percentage breakdowns, which wasn't explicitly planned in Week 4 but proved valuable for accurately communicating each factor's true predictive strength.

7. Cross-Track Collaboration Completed

Shared this week's KPI correlation findings with the Data Science track to inform their feature selection for the no-show prediction model, highlighting Booking Lead Time and Previous No-Show History as the strongest candidate features.

8. Remaining Work

Deeper investigation into combined/interaction effects (e.g., does a patient with high previous no-shows AND a long booking lead time show an even higher combined risk than either factor alone) has not yet been explored, and would strengthen the analysis further.

9. Proposed Focus for Week 6

Week 6 will focus on synthesizing this week's individual-factor findings into a more integrated view — potentially exploring combined risk segments — and translating insights into a final set of prioritized, actionable recommendations for HealthConnect leadership.